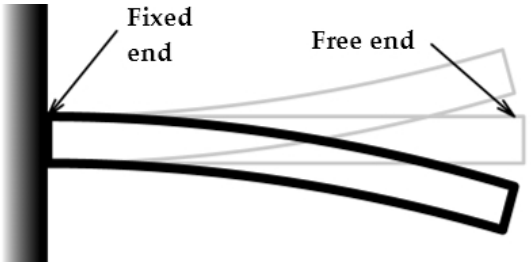
Assigned: Monday Sept. 28, 2026
Due: Monday Oct. 12, 2026

MATH 6660: Homework 3
Fall 2026

Beam Equation

Consider the Beam equation from mechanics with boundary conditions that model a *cantilever* beam:

$$\begin{aligned} \textbf{PDE}: &\quad u^{(iv)} = f(x) \quad x \in (0,1)\,, \\ \textbf{BC}: &\quad u(0) = 0, \quad u'(0) = 0, \quad u''(1) = 0, \quad u'''(1) = 0. \end{aligned}$$



1. Recast this problem as a variational problem. Clearly state the test and trial function spaces.
2. Interpret the variational problem as an energy minimization problem, clearly stating the energy functional. Prove that the variational problem and the energy minimization problems are equivalent.
3. Develop a CG(3) finite element method for this problem.
4. Prove an *a priori* error estimate for this method in the *energy norm*:

$$\|e\|_E = \left\{ \int_0^1 \left(e''\right)^2 \, dx \right\}^{1/2}.$$

5. Prove an *a priori* error estimate for this method in the L_2 norm:

$$\|e\|_{L^2} = \left\{ \int_0^1 (e)^2 \, dx \right\}^{1/2}.$$

6. Numerically implement the cG(3) method on a uniform mesh with the following functions

$$\begin{aligned} f(x) &= p_1(x)\,\cos(2\pi x) + p_2(x)\,\sin(2\pi x), \\ p_1(x) &= -64\pi(18 - 6\pi^4 x^2 + 18x^3\pi^2 + 2\pi^4 x - 36\pi^2 x^2 - 27x + 12\pi^2 x + 4\pi^4 x^3 + 3\pi^2), \\ p_2(x) &= 216 - 816\pi^2 - 96\pi^4 + 32\pi^6 x^2 - 384\pi^4 x^3 - 64\pi^6 x^3 - 288\pi^4 x^2 \\ &\quad + 3456\pi^2 x + 576\pi^4 x - 2592\pi^2 x^2 + 144\pi^4 x^4 + 32\pi^6 x^4, \\ u(x) &= \left((18 + 2\pi^2)x^2 + (-24 - 4\pi^2)x^3 + (9 + 2\pi^2)x^4\right)\sin(2\pi x). \end{aligned}$$

Numerically determine the convergence rate in both the *energy norm* and the L^2-norm by creating an appropriate convergence table.